

Venue Audio Monitor

ELEG298 S26 IOT Device

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Design Issues/Goals

Goals:

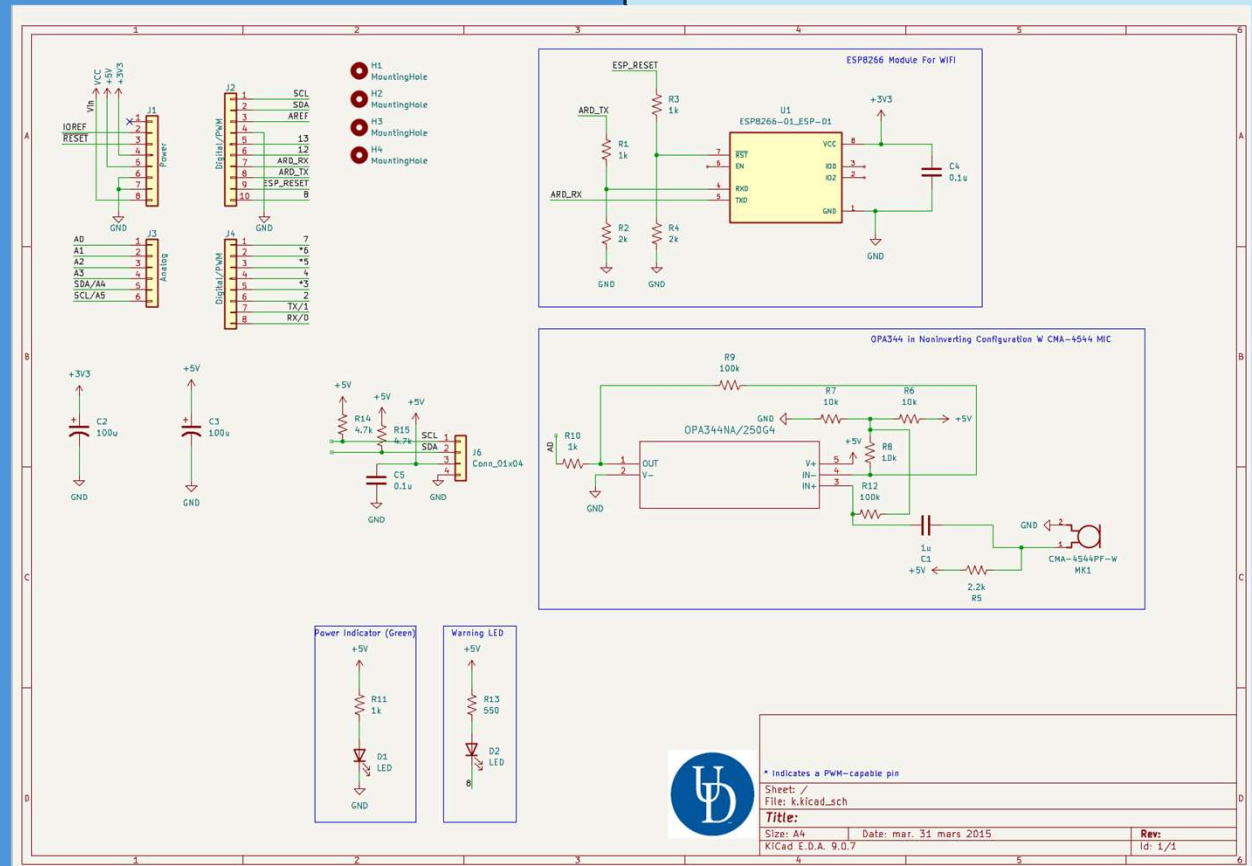
- **Design and Develop a venue audio monitor to comply with city ordinance on volume**
- **Have expandable ports for future needs**
- **Upload via ESP8266 and show with a red LED when dB volume exceed 70+**

Issue:

When designing the PCB I mistaken connected the indicator LED directly to 5V leading it to be unable to send a warning flash when too loud. However, as a prototype I designed the LED in a way so that removing R13 and wiring the LED straight to the analog pinout for the mic (A0) to have full functionality.

Schematic and Description

The IoT shield takes a audio impute via the microphone and boosts the AC signal into a operational amplifier wired in non-inverting bias to create the maximum AC output. This is then taken into the ADC to be processed as a digital signal for the ESP8266 to then take the average of peak-to-peak waves for the conversion to dB for accurate readings. The direct AC signal is also put into the indicator LED to show when the dB exceeds threshold value.



Operational Amplifier/Microphone Datasheet

resistive load in parallel with the capacitive load, a voltage divider is created, introducing a dc error at the output and slightly reducing the output swing. The error introduced is proportional to the ratio R_2/R_1 , and is generally negligible.

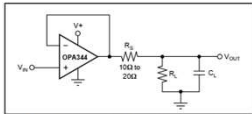


FIGURE 5. Series Resistor in Unity-Gain Configuration Improves Capacitive Load Drive.

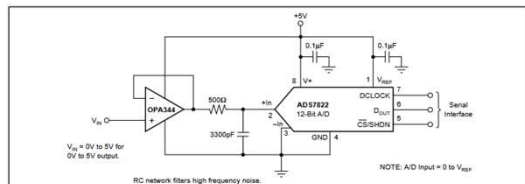


FIGURE 6. OPA344 in Noninverting Configuration Driving ADS7822.

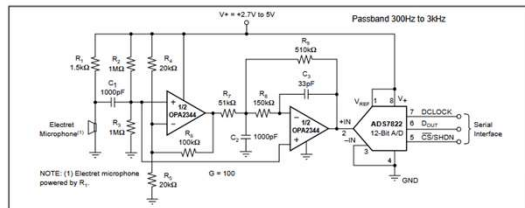


FIGURE 7. Speech Bandpass Filtered Data Acquisition System.

Figures 6 shows the OPA344 in a basic noninverting configuration driving the ADS7822. The ADS7822 is a 12-bit, micro-power sampling converter in the MSOP-8 package. When used with the low-power, miniature packages of the OPA344, the combination is ideal for space-limited, low-power applications. In this configuration, an RC network at the A/D's input can be used to filter charge injection.

Figure 7 shows the OPA344 driving an ADS7822 in a speech bandpass filtered data acquisition system. This small, low-cost solution provides the necessary amplification and signal conditioning to interface directly with an electret microphone. This circuit will operate with $V_S = +2.7V$ to $+5V$ with less than 500µA quiescent current.



OPA344
OPA2344
OPA4344
OPA345
OPA2345
OPA4345

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LOW POWER, SINGLE-SUPPLY, RAIL-TO-RAIL OPERATIONAL AMPLIFIERS MicroAmplifier™ Series

FEATURES

- RAIL-TO-RAIL INPUT
- RAIL-TO-RAIL OUTPUT (within 1mV)
- LOW QUIESCENT CURRENT: 150µA typ
- MicroSIZE PACKAGES
 - SO-8
 - MSOP-8
 - TSSOP-14
- GAIN-BANDWIDTH
 - OPA344: 1MHz, $G \geq 1$
 - OPA345: 3MHz, $G \geq 5$
- SLEW RATE
 - OPA344: 0.8V/µs
 - OPA345: 2V/µs
- THD + NOISE: 0.006%

APPLICATIONS

- PCMCIA CARDS
- DATA ACQUISITION
- PROCESS CONTROL
- AUDIO PROCESSING
- COMMUNICATIONS
- ACTIVE FILTERS
- TEST EQUIPMENT

DESCRIPTION

The OPA344 and OPA345 series rail-to-rail CMOS operational amplifiers are designed for precision, low-power, miniature applications. The OPA344 is unity gain stable, while the OPA345 is optimized for gains greater than or equal to five, and has a gain-bandwidth product of 3MHz.

The OPA344 and OPA345 are optimized to operate on a single supply from 2.5V and up to 5.5V with an input common-mode voltage range that extends 30mV beyond the supplies. Quiescent current is only 250µA (max).

Rail-to-rail input and output make them ideal for driving sampling analog-to-digital converters. They are also well suited for general purpose and audio applications and providing LV conversion at the output of D/A converters. Single, dual and quad versions have identical specs for design flexibility.

A variety of packages are available. All are specified for operation from $-40^{\circ}C$ to $85^{\circ}C$. A SPICE macromodel for design analysis is available for download from www.ti.com.



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date 06/2008

PART NUMBER: CMA-4544PF-W

DESCRIPTION: electret condenser microphone

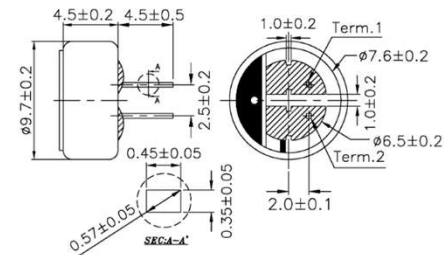
SPECIFICATIONS

directivity	omnidirectional
sensitivity (S)	-44 ±2 dB f = 1KHz, 1Pa 0dB = 1V/Pa
sensitivity reduction (ΔS-Vs)	-3 dB f = 1KHz, 1Pa Vs = 3.0 ~ 2.0 V dc
operating voltage	3 V dc (standard), 10 V dc (max.)
output impedance (Zout)	2.2 KΩ f = 1KHz, 1Pa
operating frequency (f)	20 ~ 20,000 Hz
current consumption (Iqss)	0.5 mA max. Vs = 3.0 V dc RL = 2.2KΩ
signal to noise ratio (S/N)	60 dBA f = 1KHz, 1Pa A-weighted
operating temperature	-20 ~ +70° C
storage temperature	-20 ~ +70° C
dimensions	ø9.7 x 4.5 mm
weight	0.80 g max.
material	Al
terminal	pin type (hand soldering only)
RoHS	yes

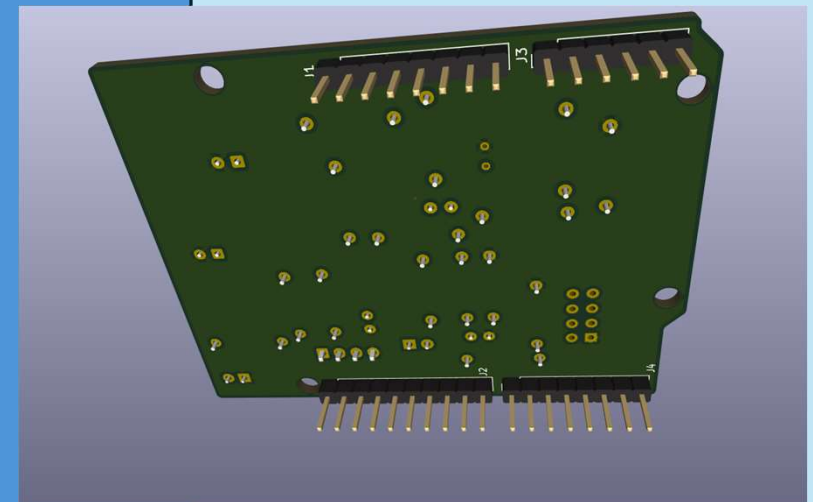
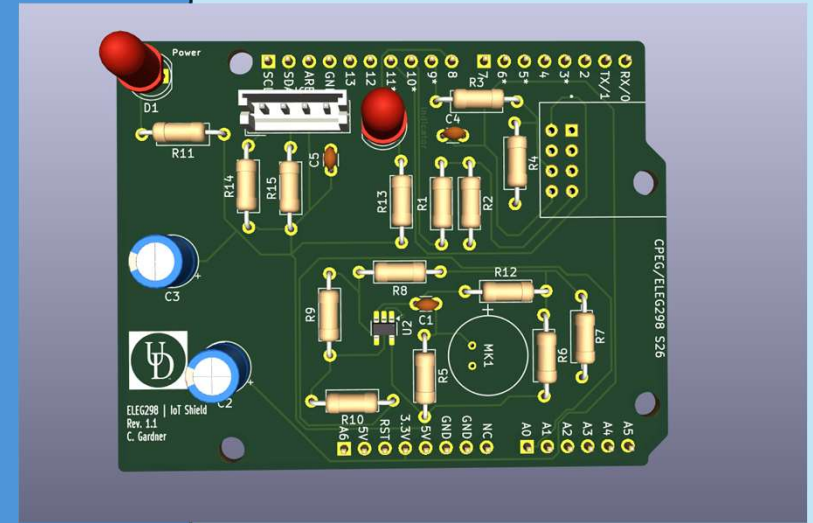
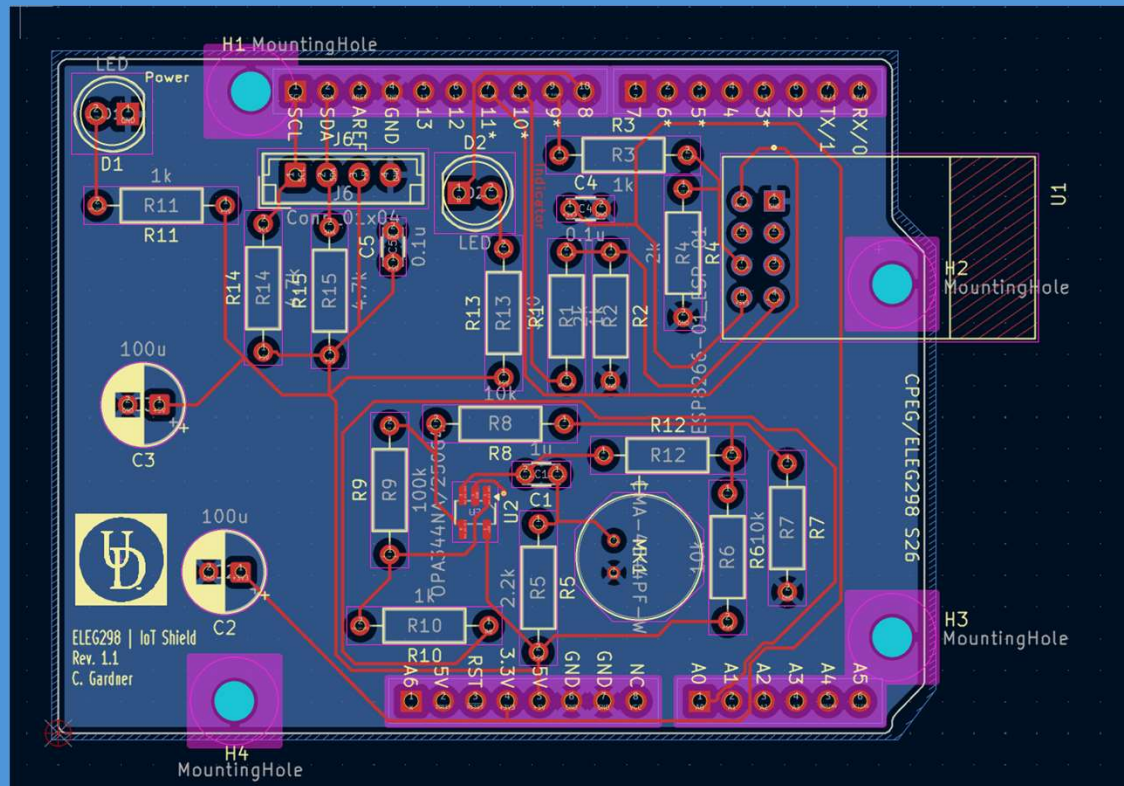
note:

We use the "Pascal (Pa)" indication of sensitivity as per the recommendation of I.E.C. (International Electrotechnical Commission). The sensitivity of "Pa" will increase 20dB compared to the "ubar" indication. Example: -60dB (0dB = 1V/Pa) ~ -40dB (1V/Pa)

APPEARANCE DRAWING



PCB Design



BOM/Stuffing Diagram

k

Rev:mar. 31 mars 2015

Ref lookupFilter

	Sour ced	Plac ed	References	Value	Footprint	Quantity
1	☐	☐	C2, C3	100u	CP_Radial_D6.3mm_P2.50mm	2
2	☐	☐	C4, C5	0.1u	C_Disc_D3.0mm_W1.6mm_P2.50mm	2
3	☐	☐	C1	1u	C_Disc_D3.0mm_W1.6mm_P2.50mm	1
4	☐	☐	R1, R3, R10, R11	1k	R_Axial_DIN0207_L6.3mm_D2.5mm_P1 0.16mm_Horizontal	4
5	☐	☐	R6, R7, R8	10k	R_Axial_DIN0207_L6.3mm_D2.5mm_P1 0.16mm_Horizontal	3
6	☐	☐	R2, R4	2k	R_Axial_DIN0207_L6.3mm_D2.5mm_P1 0.16mm_Horizontal	2
7	☐	☐	R9, R12	100k	R_Axial_DIN0207_L6.3mm_D2.5mm_P1 0.16mm_Horizontal	2
8	☐	☐	R14, R15	4.7k	R_Axial_DIN0207_L6.3mm_D2.5mm_P1 0.16mm_Horizontal	2
9	☐	☐	R5	2.2k	R_Axial_DIN0207_L6.3mm_D2.5mm_P1 0.16mm_Horizontal	1
11	☐	☐	D1, D2	LED	LED_D5.0mm	2
12	☐	☐	U1	ESP8266-01_ESP-01	XCVR_ESP8266-01_ESP-01	1
13	☐	☐	U2	OPA344NA/250G4	SOT-23-5_HandSoldering	1
14	☐	☐	MK1	CMA-4544PF-W	CMA_CUI_9P7X9P7_CUD	1
15	☐	☐	J1	Power	PinHeader_1x08_P2.54mm_Vertical	1
16	☐	☐	J2	Digital/PWM	PinHeader_1x10_P2.54mm_Vertical	1
17	☐	☐	J3	Analog	PinHeader_1x06_P2.54mm_Vertical	1
18	☐	☐	J4	Digital/PWM	PinHeader_1x08_P2.54mm_Vertical	1
19	☐	☐	J6	Conn_01x04	JST_EH_B4B-EH-A_1x04_P2.50mm_Vertical	1

Arduino Code

```
const int sensorpin = A0;

String espData(String command, const long timeout, boolean debug);

void setup() {

  Serial.begin(115200);
  espSerial.begin(9600);

  pinMode(sensorpin, INPUT);

  pinMode(RESET_PIN, OUTPUT);
  digitalWrite(RESET_PIN, LOW);
  delay(500);
  digitalWrite(RESET_PIN, HIGH);

  Serial.println("\n\nCPEG/ELEG Demo S26 rev1.0");
  Serial.println("setting up Adafruit IO");

  response = espData("get_macaddr", 2000, true);
  response = espData("get_version", 2000, true);
  response = espData("wifi_ssid=" + WIFI_SSID, 4000, true);
  response = espData("wifi_pass=" + WIFI_PASS, 4000, false);
  response = espData("io_key=" + IO_KEY, 4000, false);
  response = espData("io_user=" + IO_USERNAME, 4000, true);

  response = espData("setup_io", 30000, true);

  if (response.indexOf("connected") < 0) {
    Serial.println("\n\nAdafruit IO Connection Failed");
    while (1)
      ;
  }

  response = espData("setup_pubfeed=1,CPEG298pub", 800, true);
  response = espData("setup_subfeed=3,reset_button", 800, true);
  response = espData("setup_time,-5", 800, true);

  Serial.println("Setup complete");
```

```
void loop() {

  int minValue = 1023;
  int maxValue = 0;

  unsigned long startTime = millis();

  while (millis() - startTime < 100) {
    int reading = analogRead(sensorpin);

    if (reading < minValue) {
      minValue = reading;
    }

    if (reading > maxValue) {
      maxValue = reading;
    }
  }

  int peakToPeak = maxValue - minValue;

  Serial.print("AC peak-to-peak value is ");
  Serial.println(peakToPeak);

  response = espData("send_data=1," + String(peakToPeak), 4000, false);

  delay(5000);
}

String espData(String command, const long timeout, boolean debug) {

  char c = 0;
  response = "";

  espSerial.println(command);

  long int time = millis();

  while ((time + timeout) > millis()) {
```

```
String espData(String command, const long timeout, boolean debug) {

  char c = 0;
  response = "";

  espSerial.println(command);

  long int time = millis();

  while ((time + timeout) > millis()) {

    while (espSerial.available()) {
      c = espSerial.read();
      response += c;
    }

    if ((c == '\n') && (command != "setup_io")) break;
  }

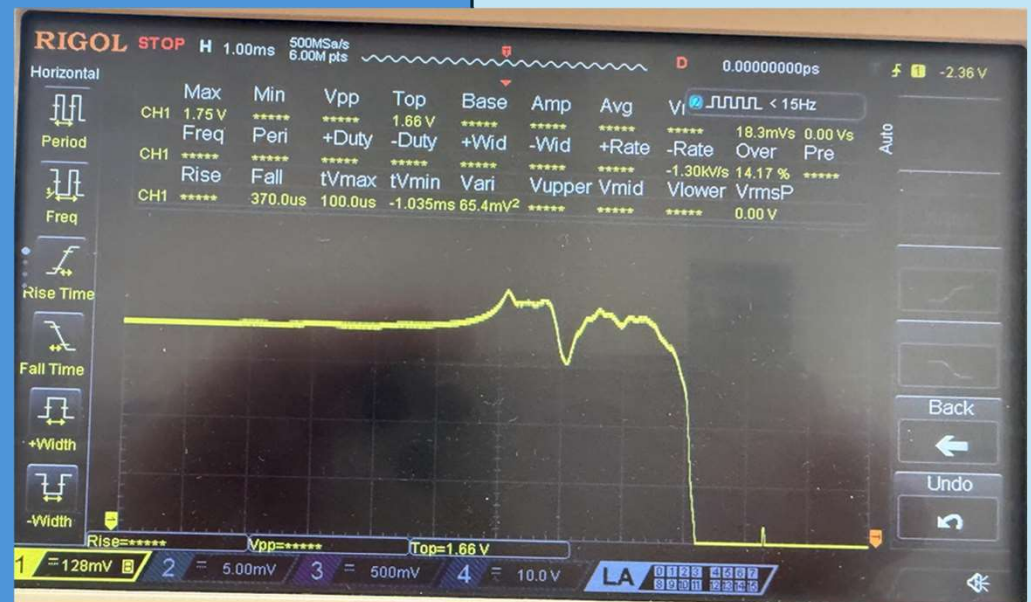
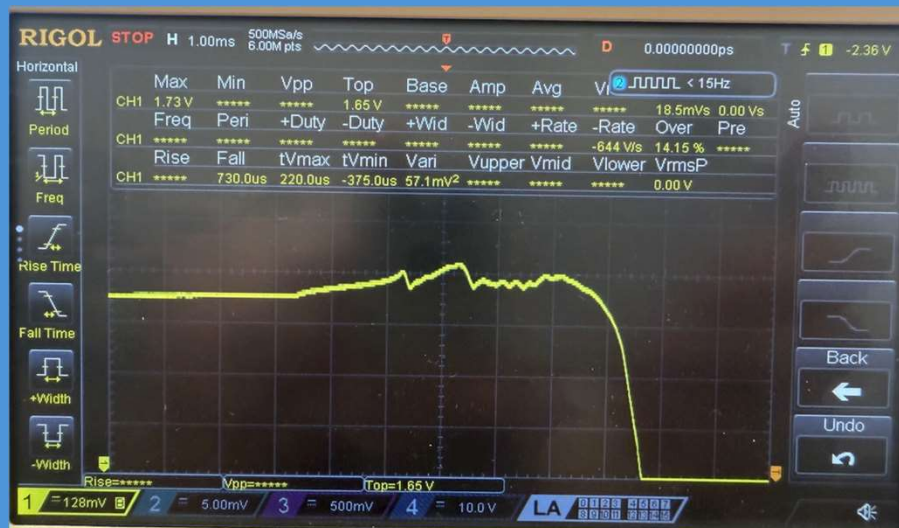
  response.trim();

  if (debug) {
    Serial.println("Resp: " + response);
  }

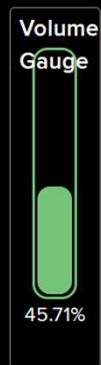
  return response;
}

Serial Monitor
```

Oscilloscope Readings from impulses



Adafruit Dashboard



Reset

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"Flexibility is the key to stability"

Demo